INFRASTRUCTURE LEDGER

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INFRA / 02
AI INFRASTRUCTURE
ECONOMICS
Capital moves first. Constraints decide what ships.
A report on capacity,
capital, and the physical stack.
A source-led editorial data study
for strategy and platform operators.
THE VALUE CHAIN IS PHYSICAL
APPLICATIONS
value
PLATFORM
distribution
COMPUTE
scarcity
POWER + PLACE
constraint
PUBLIC DISCLOSURES / ILLUSTRATIVE SYSTEMS ANALYSIS
SOURCES: ALPHABET IR · MICROSOFT IR · SEE SOURCE NOTES IN PAGE PLAN

NEUTRAL VISUAL FIELD

01 / THE STACK

AI is not one market. It is a stack of dependencies.
The margin pool can move upward; the bottleneck often stays physical.

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INFRA / 02

07 / APPLICATIONS
workflow · distribution · willingness to pay

06 / MODELS
training · inference · evaluation

05 / SOFTWARE
orchestration · serving · observability

04 / SYSTEMS
accelerator · memory · network fabric

03 / DATA CENTERS
land · cooling · power delivery

dependency ↑

INTERPRETATION / THE ECONOMICS OF AI ARE DISTRIBUTED ACROSS THE STACK

NEUTRAL VISUAL FIELD

02 / CAPITAL

Capital is accelerating — but the periods are not interchangeable.
Reported figures are shown with company and fiscal-period labels, not as a false league table.

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Alphabet FY2025
$91.4B

Microsoft FY25 Q4
$24.2B

Alphabet FY2026 outlook
$175–185B

READ THIS AS
directional evidence
not a single-period ranking
different disclosure periods
different business mix

WHAT IS REPORTED
WHAT IT SUGGESTS
WHAT IT DOES NOT PROVE

Large technical-infrastructure commitments are visible.

Capacity is becoming a strategic balance-sheet item.

CapEx alone does not tell us who captures the margin.

SOURCES: ALPHABET 2025 Q4 EARNINGS CALL · MICROSOFT FY25 Q4 EARNINGS · FIGURES IN USD

NEUTRAL VISUAL FIELD

03 / COMPOSITION

Where the money lands changes the depreciation story.
One disclosed split makes the physical stack visible.
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ALPHABET FY2025
$91.4B
technical infrastructure CapEx
APPROXIMATE 2025 COMPOSITION
60% SERVERS
40% DC + NETWORK
SHORTER CYCLE
servers
faster refresh
utilization pressure
LONGER DURATION
data centers
networking
depreciation horizon
The stack has different clocks.
A server cycle can move faster
than a building or grid connection.
SOURCE: ALPHABET 2025 Q4 EARNINGS CALL / APPROXIMATELY 60% SERVERS, 40% DATA CENTERS + NETWORKING

NEUTRAL VISUAL FIELD

04 / BOTTLENECK

The constraint is not “compute.” It is the coordination of four physical systems.
Land, energy, networking, and servers arrive on different timelines.

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LAND

permitted + buildable

lead time ≠ demand

ENERGY

predictable supply

lead time ≠ demand

NETWORK

fabric + interconnect

lead time ≠ demand

SERVERS

GPU + components

lead time ≠ demand

CAPACITY ONLY SHIPS WHEN THE SLOWEST LAYER ARRIVES.

This is why supply-chain risk belongs inside the AI strategy — not in a footnote after the model roadmap.

SOURCE: MICROSOFT FY2025 ANNUAL REPORT / DATA CENTER LAND, ENERGY, NETWORKING, SERVERS

NEUTRAL VISUAL FIELD

05 / VALUE

Capacity becomes economics only through utilization.
The flywheel is simple to draw — and difficult to keep turning.
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OPERATING
LEVERAGE
CAPACITY
hardware + power
REVENUE
serving demand
CASH
depreciation + margin
UTILIZATION
load + scheduling
NO UTILIZATION → NO LEVERAGE
INTERPRETATION / THE UNIT ECONOMICS OF AI ARE A CAPACITY-MANAGEMENT PROBLEM

NEUTRAL VISUAL FIELD

06 / ECONOMICS

Training is episodic. Inference is continuous.
Illustrative curves show why utilization and serving efficiency can matter more than a single benchmark.
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INFRA / 02
COST
TIME
TRAINING / burst
INFERENCE / stream
OPERATOR QUESTION
Can the load be
scheduled, shifted,
or pooled?
ILLUSTRATIVE ONLY
Shape of curve
is conceptual
not a benchmark
ANALYTICAL MODEL / ILLUSTRATIVE CURVES — NOT A PERFORMANCE CLAIM

NEUTRAL VISUAL FIELD

07 / MARGIN

The margin pool is a flow, not a single layer.
Compute creates the possibility; platforms and applications decide where value compounds.

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POWER + PLACE

fixed assets

COMPUTE

scarce capacity

PLATFORM

orchestration

APPLICATION

workflow value

CAPITAL INTENSITY

OPERATING LEVERAGE

CUSTOMER LOCK-IN

low flexibility

high fixed cost

shared capacity

better scheduling

workflow fit

switching friction

INTERPRETATION / VALUE MOVES UP THE STACK, BUT DEPENDENCIES MOVE WITH IT

NEUTRAL VISUAL FIELD

08 / RISK

The dangerous asset is capacity you cannot use.
A simple matrix makes the hidden risk visible: high lead time plus low utilization.
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INFRA / 02
UTILIZATION →
L
E
A
D
T
I
M
E
UNDERUSE
capacity exists
demand is soft
HEALTHY
matched load
optionality
STRANDED
fixed cost
no flexibility
BOTTLENECK
demand exists
supply lags
WATCH ITEM
Lead time
can outlive
demand.
Treat capacity as an option

NEUTRAL VISUAL FIELD

09 / CHOICE

There are three ways to buy capacity. None is free.
The right answer depends on demand certainty, capital appetite, and control requirements.

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BUY
control
RENT
optionality
BUILD
differentiation
CAPITAL
high
variable
very high
SPEED
medium
fast
slow
CONTROL
high
shared
highest
RISK
demand
vendor
execution
OPERATOR TEST
How certain
is demand?
How unique

NEUTRAL VISUAL FIELD

10 / CONTROL

Run infrastructure like an operating system, not a warehouse.
Five measures reveal whether the asset is earning its keep.

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CAPACITY
what exists
01

UTILIZATION
what is used
02

LATENCY
what users feel
03

DEPRECIATION
what time costs
04

POWER
what scale requires
05

THE SCORECARD TURNS CAPEX INTO A FEEDBACK SYSTEM.

If one measure is missing, the operator is managing a blind spot — not a platform.

OPERATOR CONTROL / MEASURE CAPACITY, UTILIZATION, USER EXPERIENCE, TIME, AND POWER TOGETHER

NEUTRAL VISUAL FIELD

11 / CLOSE

The winning infrastructure is not the biggest. It is the most legible.
Capital creates the option. Constraints define the clock. Control decides the return.
12 / 12
INFRA / 02
Capital creates
the option.
Constraints define
the clock.
Control decides
the return.
THE OPERATOR’S QUESTION
Where is the bottleneck?
What can we flex?
What must we own?
MAKE THE STACK LEGIBLE.
AI INFRASTRUCTURE ECONOMICS / END OF REPORT

NEUTRAL VISUAL FIELD